

human-like text, and are augmented with the ability to execute plans involving multiple steps. The agent architecture comprises:

- **Natural language understanding (NLU):** Utilises LLMs to parse and comprehend user requests.
- **Planning mechanism:** Generates a sequence of actions to fulfil the user's intent.
- **Execution engine:** Orchestrates API calls and manages the workflow execution.
- **Memory and state management:** Maintains context over multiple interactions to ensure coherent dialogue and task execution.

Multi-step task automation in banking with Amazon Bedrock Agents

Banking operations often involve intricate processes requiring interaction with multiple systems, verification steps, and compliance checks. Amazon Bedrock Agents can automate these multi-step tasks by:

- **Interfacing with core banking systems:** Through secure API integrations, agents can access account information, transaction histories, and customer data.
- **Performing data validation and verification:** Agents can validate inputs against business rules, ensuring data integrity and compliance.
- **Executing conditional logic:** Based on intermediary results, agents can make decisions, trigger alerts, or escalate issues.
- **Maintaining transactional integrity:** Agents can support ACID (atomicity, consistency, isolation, durability) properties by leveraging transactional APIs.

Here are some use cases.

- **Loan application processing**
Step 1: The agent collects applicant information via a conversational interface.

Step 2: It validates the information against eligibility criteria using internal APIs.

Step 3: Performs credit scoring by interfacing with third-party services.

Step 4: Generates a risk assessment report.

Step 5: If approved, initiates the loan issuance process; if not, provides feedback and alternatives to the applicant.

- **Fraud detection and response**

Step 1: Monitors transaction patterns in real-time.

Step 2: Identifies anomalies using machine learning models.

Step 3: Validates suspicious activities by cross-referencing with user behaviour data.

Step 4: Engages with the customer for verification via secure messaging.

Step 5: Takes action such as blocking transactions or accounts and escalates to compliance teams if necessary.

- **Regulatory compliance reporting**

Step 1: Aggregates data from various banking systems.

Step 2: Validates data against regulatory requirements.

Step 3: Compiles reports in required formats.

Step 4: Submits reports to regulatory bodies via secure channels.

Step 5: Archives reports and updates

compliance logs.

Future trends

Amazon Bedrock is designed to support future trends and opportunities in the financial industry.

- **Digital transformation:** It enables customers to transform their business models, processes, and offerings, using digital technologies and channels.
- **Open banking:** Customers can participate and benefit from the open banking ecosystem, which allows them to access and share their financial data and services across different providers and platforms.
- **Artificial intelligence:** It helps customers to leverage the power and potential of artificial intelligence, which enables machines to perform tasks that require human intelligence, such as perception, understanding, learning, and decision making.
- **Blockchain:** Amazon Bedrock enables customers to explore and experiment with blockchain.
- **Cloud-native development:** It helps customers to adopt and embrace cloud-native development, which is a software development approach that leverages cloud capabilities and characteristics, such as scalability, elasticity, resilience, and automation. **END** 

References

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